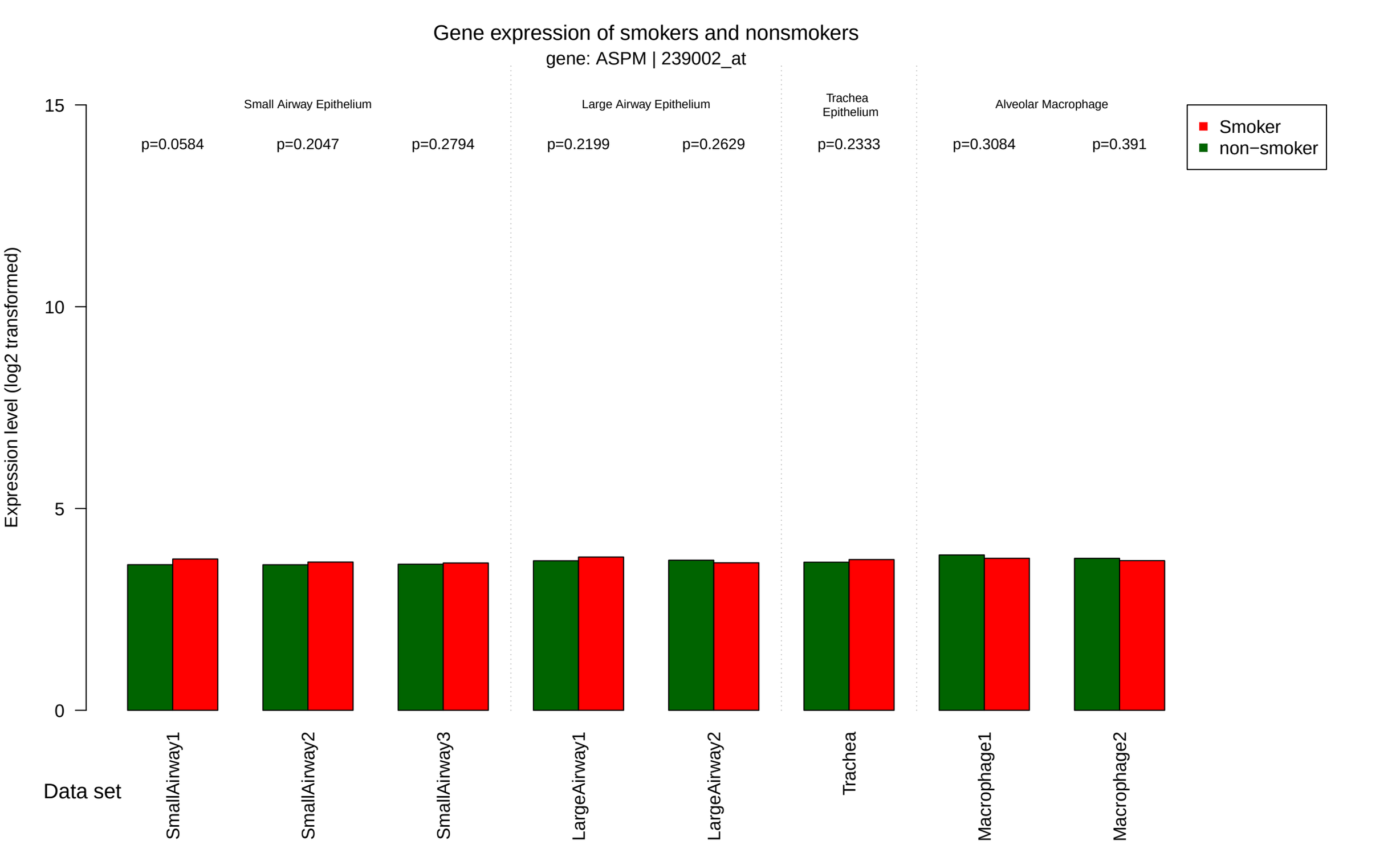
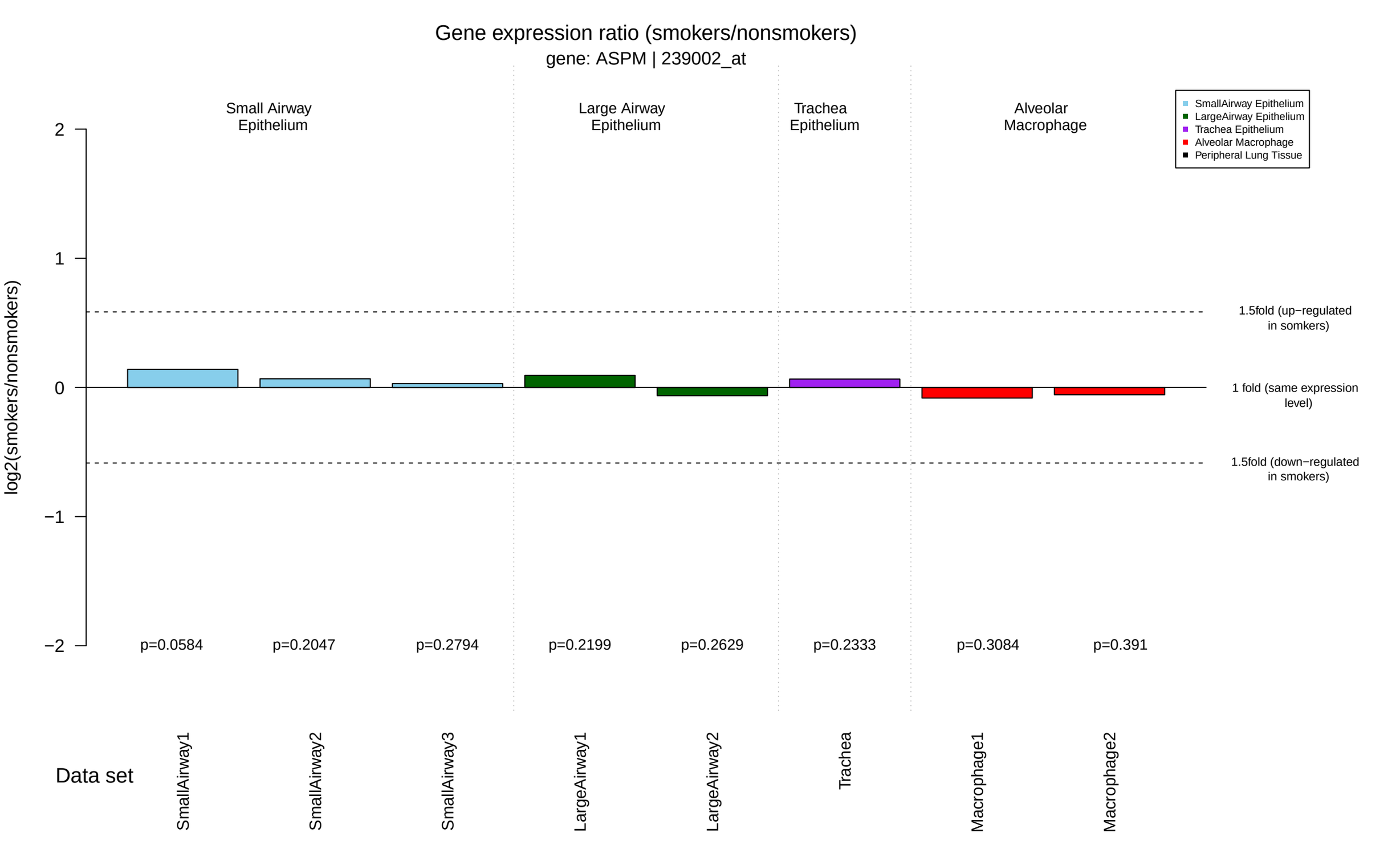
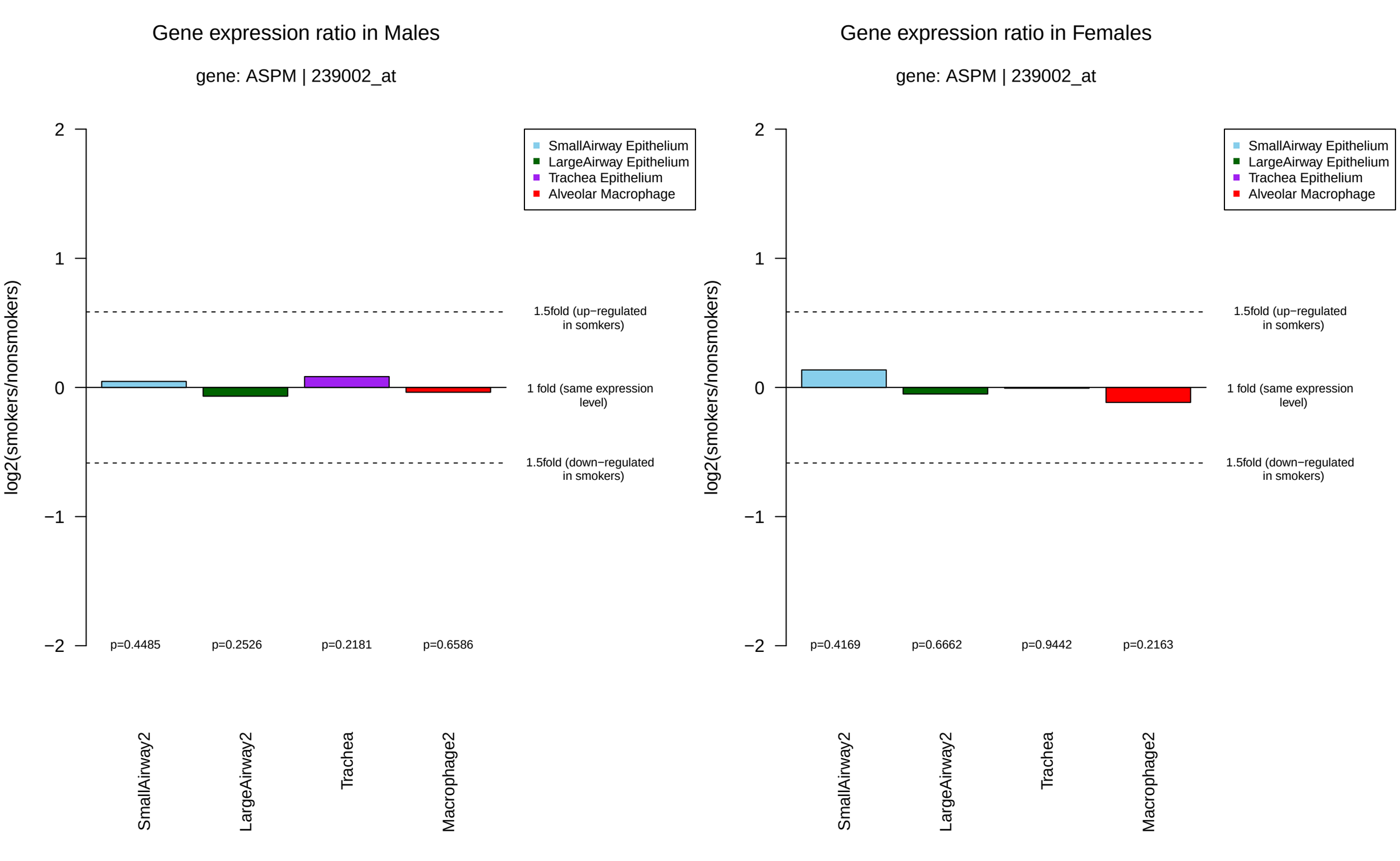
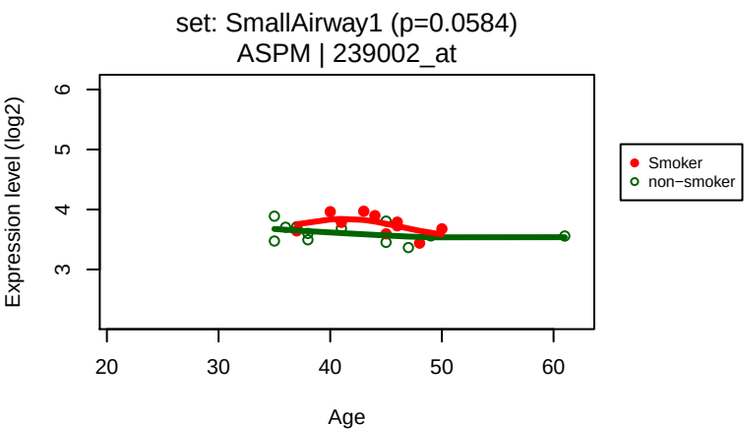
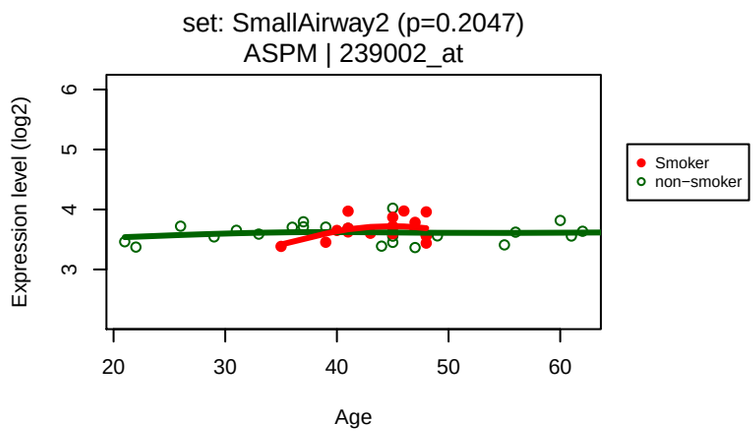
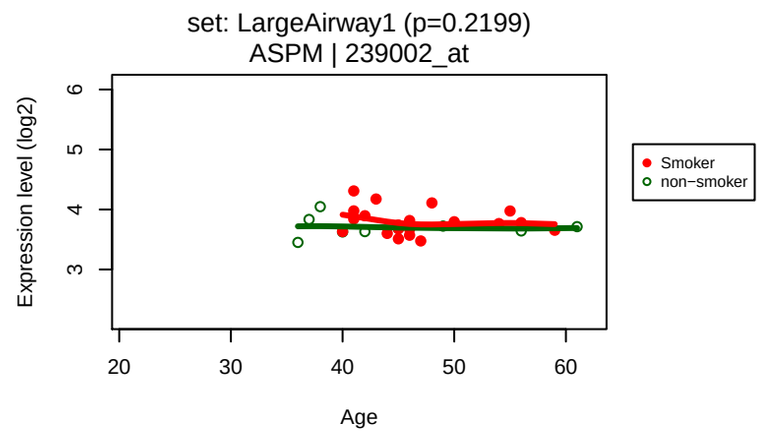

Gene expression of smokers and nonsmokers
gene: ASPM | 239002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0584
p=0.2047
p=0.2794
p=0.2199
p=0.2629
p=0.2333
p=0.3084
p=0.391
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

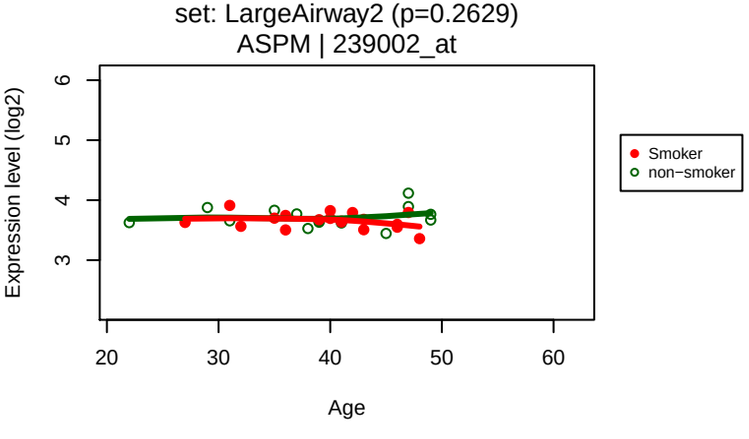

Gene expression ratio (smokers/nonsmokers)
gene: ASPM | 239002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0584
p=0.2047
p=0.2794
p=0.2199
p=0.2629
p=0.2333
p=0.3084
p=0.391
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

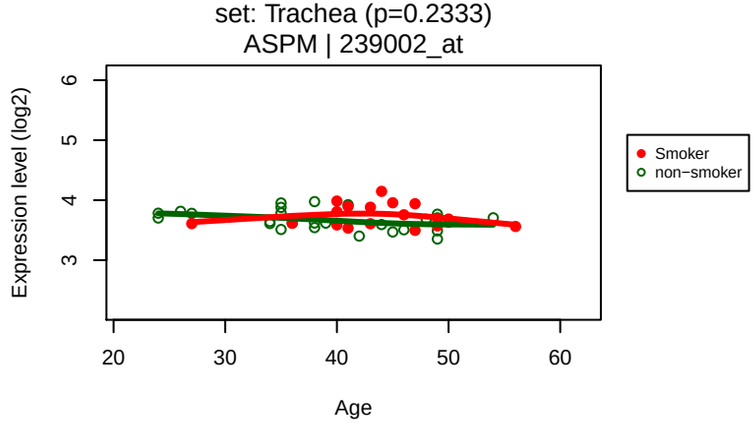

Gene expression ratio in Males
gene: ASPM | 239002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4485
p=0.2526
p=0.2181
p=0.6586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPM | 239002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4169
p=0.6662
p=0.9442
p=0.2163
SmallAirway2
LargeAirway2
Trachea
Macrophage2

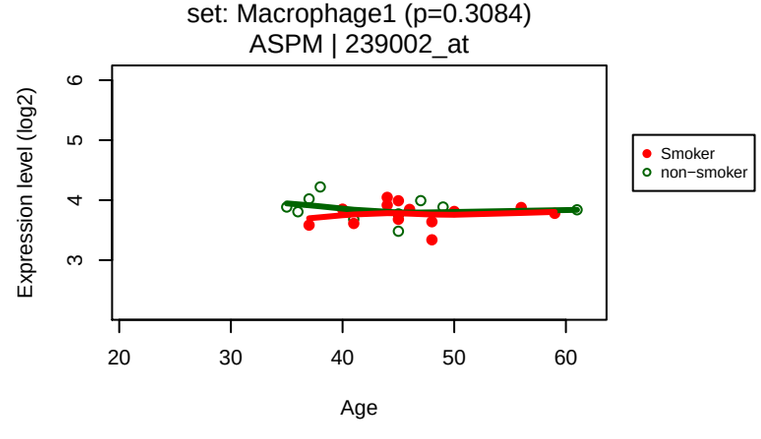

set: SmallAirway1 (p=0.0584)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker

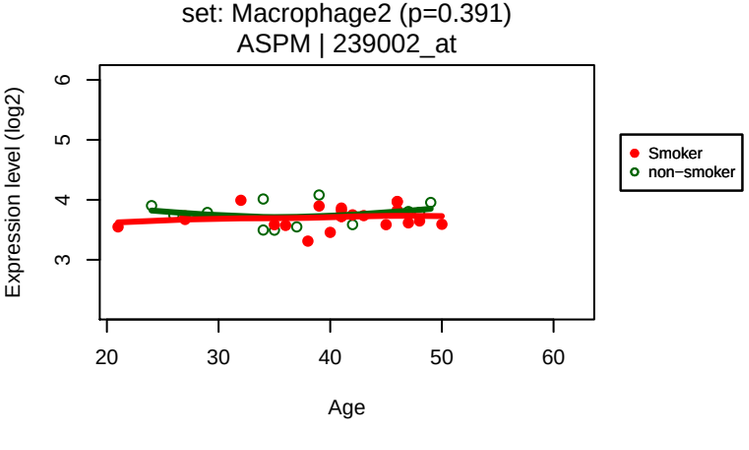

set: SmallAirway2 (p=0.2047)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker

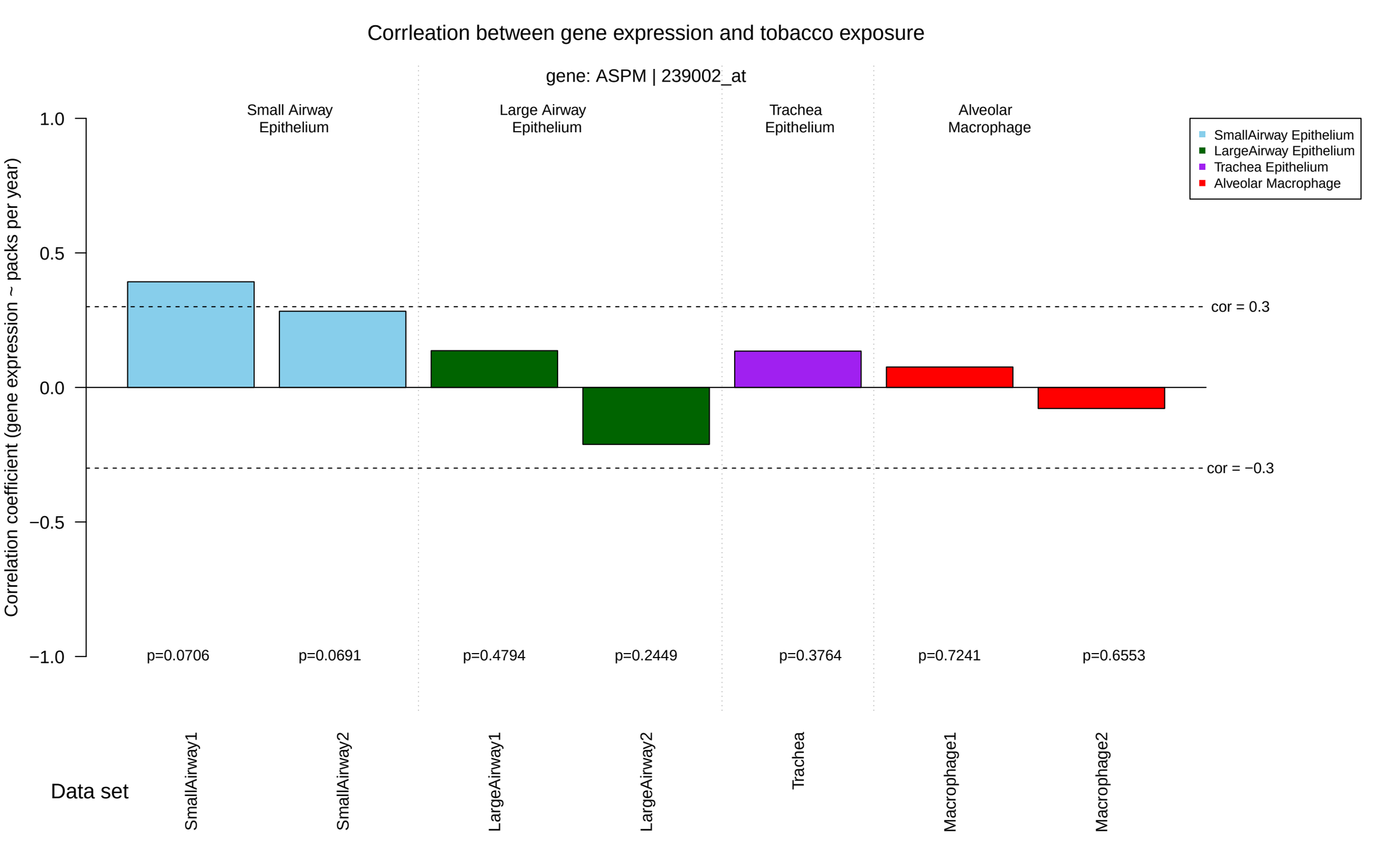

set: LargeAirway1 (p=0.2199)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2629)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2333)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3084)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.391)
ASPM | 239002_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPM | 239002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0706
p=0.0691
p=0.4794
p=0.2449
p=0.3764
p=0.7241
p=0.6553
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2